

The examination is being carried out on the **following application documents**

Description, Pages

1-305 as published

Claims, Numbers

1-15 filed in electronic form on 24-08-2023

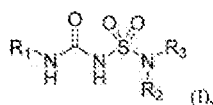
- 1 With the letter of 24.08.2023, the Applicant filed a new set of claims and arguments for supporting the patentability of the claimed subject-matter. After considering the amendments from the current set of claims and the arguments provided during the whole written procedure, the Examining Division is still of the opinion that the EPC requirements are not fulfilled.

As requested by the above-mentioned letter the Applicant is invited to attend oral proceedings (Article 116 EPC) to come to a final decision regarding the patentability of the present application. The oral proceedings are to be held by video-conference in accordance with Article 1(1) of the decision of the President of the EPO dated 1 April 2020 and the members of the Examining Division may connect to the oral proceedings from different locations (see Article 2(2) of the aforementioned decision). The rules for oral proceedings held by video-conference apply (see the Guidelines E-III, 11). During pauses for deliberation, the members of the examining division may deliberate and vote among themselves by separate videoconference.

- 2 The main aspects to be discussed during the oral proceedings are:

2.1 Inventive step (Article 84 EPC)

The present application claims compounds of formula (I):



wherein:

- R₁ is a cycloalkyl or a bi or tri substituted phenyl,

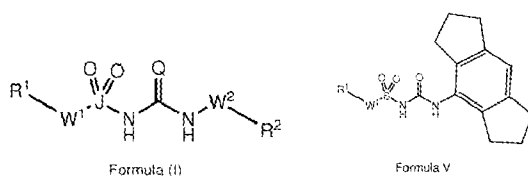
-R₂ is -(CX₁X₂)_n-R_{2s} with n=0-2 and R_{2s}=heterocycloalkyl

-R₃ is a heterocycloalkyl or heterocycloaryl

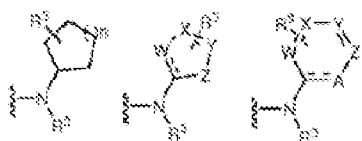
useful to treat diseases connected to the inhibition of the NLRP3 activation. In particular, the compounds were tested for their inhibitory activity against IL-1 beta release upon NLRP3 activation in peripheral blood mononuclear cells (PBMC).

Closest prior art

D1 (WO 2017/140778), which is regarded as being the closest prior art, discloses compounds of formula (I) (claim 1) or (V)(claim 7) with the same uses as the present ones.

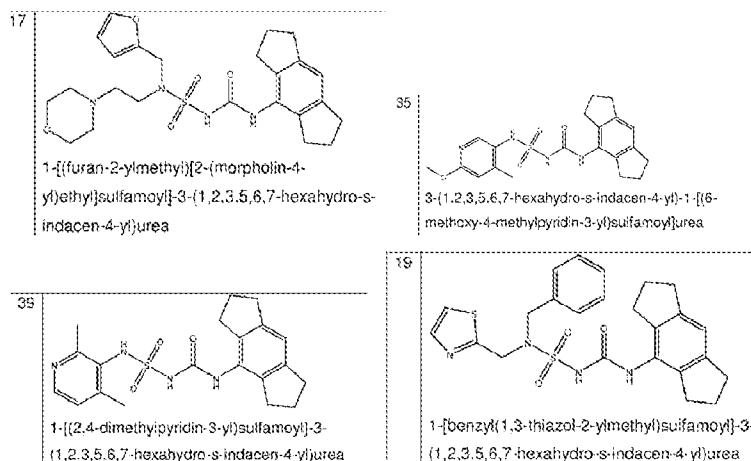


The subject-matter of current claim 1 can be considered as a selection from the compounds of formula I or V disclosed in D1 (claims 1 and 7). For example starting from the compounds of formula (I) of D1 for arriving at the present claimed compounds it has to be selected J to be S, W to be -N (R₁)₂ (e.g. claims 2, 5 of D1), R₁ to be an aryl or heteroalkyl, Q=O and W₂R₂ a ring. Further in order to arrive from the compounds of formula (V) of D1 at the present claimed ones, then the single selection to be done is the substituent R₁W₁, which has to be one of the following preferred groups disclosed in claim 6 of D1:



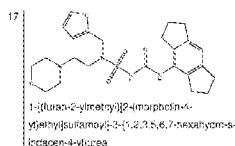
wherein this moiety is a -N(heterocycle)R₃ structure and R₃ is a heterocyclyl or heteroaryl.

As regards the specific examples disclosed in D1, compounds **17**, **19**, **35** and **39** appear to be the closest structural ones.

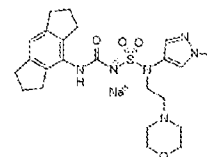


With the letter of 07.10.2022, the Applicant filed comparative data between

compound 17 of D1



and present example 46:



Example 17 of D1 has in the PBMC assay an IC₅₀ higher than 10µm while the present example 46 has an IC₅₀ lower than 1µm.

The two tested compounds differ between them through the fact that the compound of D1 contains a methylfuranly moiety instead of the present N-methylpyrazol (example 17 of D1 does not contain a directly linked heterocycle (mandatory for the present R3) and additionally the nature of this heteroaryl ring is different (furan instead of pyrazol). Moreover, the two compared examples differ also through the fact that the present compound is tested as sodium salt while the one of D1 as free base (the sodium salt could also have an impact on the claimed activity).

Technical effect and problem to be solved

In view of the comparative data provided by the Applicant and the fact that the present compounds represent a novel selection from the general structures disclosed in D1, the technical problem underlying the present application is to be seen in the provision of further sulfonureas derivatives useful to treat diseases connected to inhibition of NLRP3 activation and having unexpected effects over the closest structural compounds disclosed in the prior art.

Solution

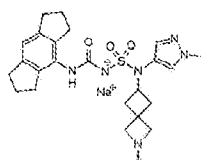
The specific combination of the present substituents R2 and R3 and the form of sodium salt leading for some of the present compounds to an improved activity re the compounds disclosed in D1.

The obviousness of the solution

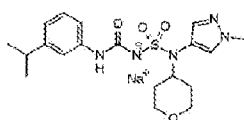
D1 compounds are disclosed to have, depending on their structure, an inhibitory activity ranging between less than 1 μm to less than 50 μm (the same range as in the present case).

Looking to the specific data provided on pages 301 to 116 of the present specification, it can be concluded that the level of activity appears to be highly dependent on the specific structure of the compound, e.g:

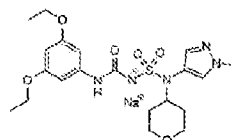
a) compounds 58, 71 and 90 have an IC₅₀ in the range of 10 to 50 μm :



comp 58

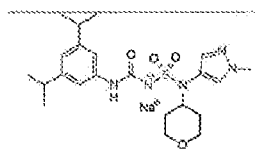


comp 71

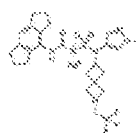


comp 90

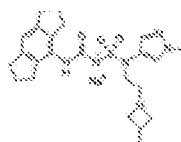
b) present examples 73, 69 and 74 have an IC₅₀ in the range between 3 and 10 μm :



comp 73

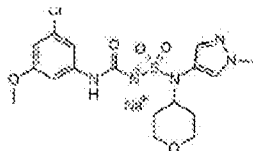


comp 69



comp 74

c) compound 93 has an IC₅₀ less than 1 μm :



At the first glance comparing the activity reported for the present example 46 and example 17 of D1 and disregarding the fact that they differ in more than one structural feature (present compound was tested as a sodium salt and not as free base and the nature of the 5-membered heteroaryl is different) an improvement of 10 times of the IC₅₀ might be considered unexpected.

The question to be answer is if the claimed breadth could be seen as a fair generalisation over the comparison data provided with the letter of 07.10.2023.

In view of the biological data provided in the application it appears not to be credible that any compound encompassed by claim 1 (having the specific combination of substituents R/R3) would possess an IC₅₀ less than 1 µm. For example compound 74 contains the same combination of substituents R₂/R₃ as the one on which an inventive step was acknowledged for the present example 46 (over example 17 of D1), but its IC₅₀ lies in the range of 3 to 10 µm, which could be considered to be close to the one reported for example 17 of D1.

Moreover, the application discloses compounds having the same NR₂R₃ moiety and wherein substituents placed in different parts of the molecule could have an important impact on the claimed activity. Compounds 58 and 21 differ only on the nature of the substituent on the nitrogen group of the azaspiro[3.3]heptan-6-yl, but their inhibitory activities are very different (in the range of 10 to 50 µm for example 58 and 3 to 10 µm for example 21); the same applies for examples 71, 90, 73 and 93 which differ among them only on the nature of the second substituent on the phenyl ring but their inhibitory activity is at completely different levels.

Consequently, considering all the biological data provided in the present application in table A and the ones provided for the same type of compounds in table 4 of D1 it cannot be concluded that all the compounds encompassed by the current claim 1 would possess the same level of improvement over the closest structural compound disclosed in D1 (a 10 times improvement). In addition, since it is general knowledge that the salts are in general more soluble than the corresponding free acids, it is expected that an improvement in solubility is connected to an improved inhibitory activity.

In view of these facts it is considered that the above technical problem was not solved over the whole claimed scope. Consequently, it has to be reformulated as the provision of further sulfamoylurea derivatives useful to treat diseases connected to the inhibition of NLRP3. Since the present compounds of claim 1 represent a novel selection from the general structures (I) and (V) of D1, the skilled person starting from compounds of D1 (in particular examples 35, 39 and 17 and 19) would arrive at the present compounds with the expectation that the same level of activity is maintained.

Thus, in order to fulfil the requirements of Article 56 EPC either experimental data will be provided supporting that substantially all the claimed compounds have an unexpected property or improved activity over the structurally closest prior art compounds (D1), which is attributable to the distinguishing feature of the invention or the scope of claim 1 will be restricted to the very specific compounds which possess indeed the claimed level of improvement.

2.2 Article 84 EPC

2.2.1 Some of the definitions provided in the claims cover structures which are not implicitly covered by the general accepted definitions of such groups. For example groups as alkyl, cycloalkyl are defined in the specification to cover also the partially unsaturated structures which normally the skilled person would not consider as falling under the terms alkyl/cycloalkyl. Thus, in view of these discrepancies the requirements of Article 84 EPC are considered not to be fulfilled.

2.2.2 The requirements of Article 84 EPC are in general not fulfilled when amended claims are filled without the corresponding adapted description.

3 At the Oral Proceedings a final decision will be taken.

The Applicant is informed that failing to provide suitable amendments such as to overcome the above-mentioned objections, will most probably result in the refusal of the application under Article 97 EPC.

Should the Applicant submit application documents which comply with the provisions of the EPC at the latest one month before the oral proceedings (Rule 116 EPC), the procedure could be continued in writing.